FINAL PROJECT | INDUSTRY-SIMULATED AI PRODUCT WORKFLOW

Will this order arrive late?

Predicting late deliveries at checkout with a CrewAI Flow: a Data Analyst crew hands off to a Data Scientist crew

Olist Brazilian e-commerce data | CrewAI + Gemini, scikit-learn, Streamlit, Flask, GitHub
github.com/matnael332/olist-crewai-flow

THE BUSINESS PROBLEM

Late deliveries destroy customer satisfaction

62.4%

of LATE orders get a 1-2 star review

9.2%

of ON-TIME orders get a 1-2 star review

6.8%

of 96,203 delivered orders arrived late



Goal: flag risky orders at checkout so operations can act before the customer is let down.

DATASET

Real marketplace data, cleaned and versioned

Source: Olist Brazilian E-Commerce, 9 tables (orders, items, payments, reviews, customers, products, sellers, geolocation, categories), pinned commit + sha256 manifest

Scale: 99,441 orders, 2016-09 to 2018-10

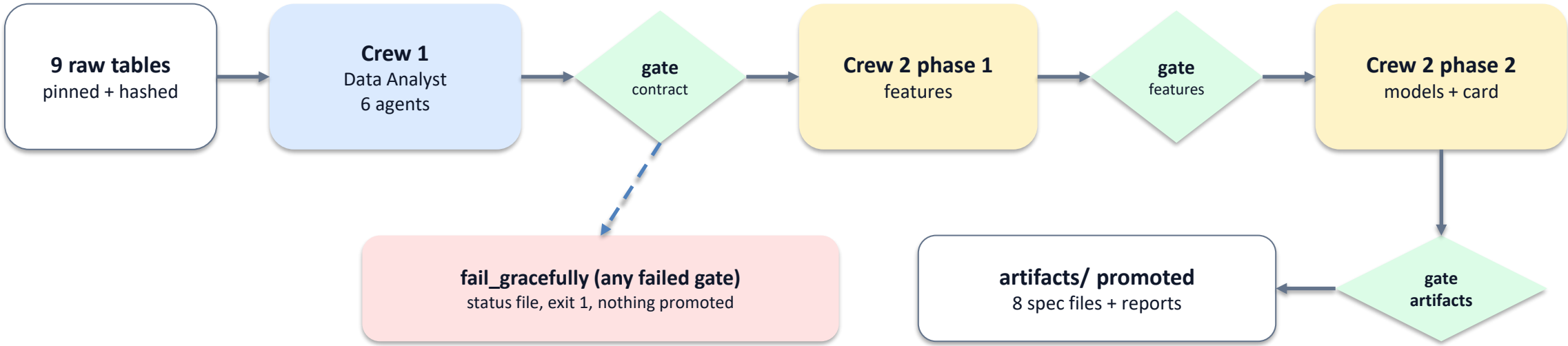
Real quality issues: non-delivered orders, missing delivery dates, out-of-sequence timestamps, products without category, duplicate reviews, invalid payments

Result: clean_data.csv with 96,203 delivered orders, one row per order, 38 documented columns

rule	rows removed
R1 keep delivered	2,963
R2 require delivery date	8
R3 purchase window	267
rows kept	**96,203**

ARCHITECTURE

Two AI crews, one Flow, three quality gates



Crew 1: data engineer, documentation writer, analytics lead, insights writer, data steward, contract author.
Crew 2: feature engineer, feature reviewer, ML engineer, model reviewer, model card author. The contract version and hash travel with the handoff.

CREW 1 INSIGHTS

What happened: fast growth and a Black Friday spike



All 99,441 orders by purchase month, every status: 97.0% were delivered. November 2017, the Black Friday month, is the tallest bar.

800 to 7,544

orders per month, Jan 2017 to Nov 2017

1,176

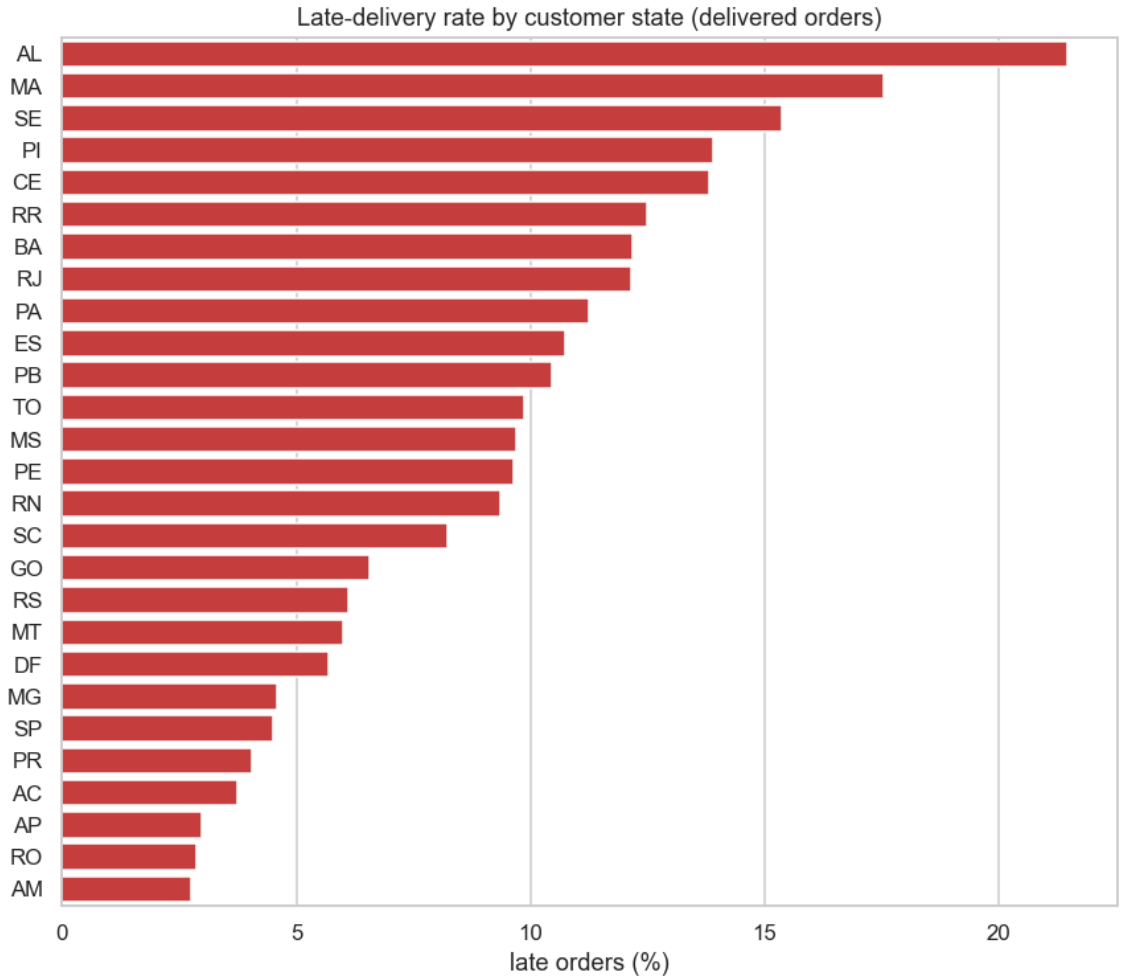
orders on 2017-11-24 (next busiest day: 499)

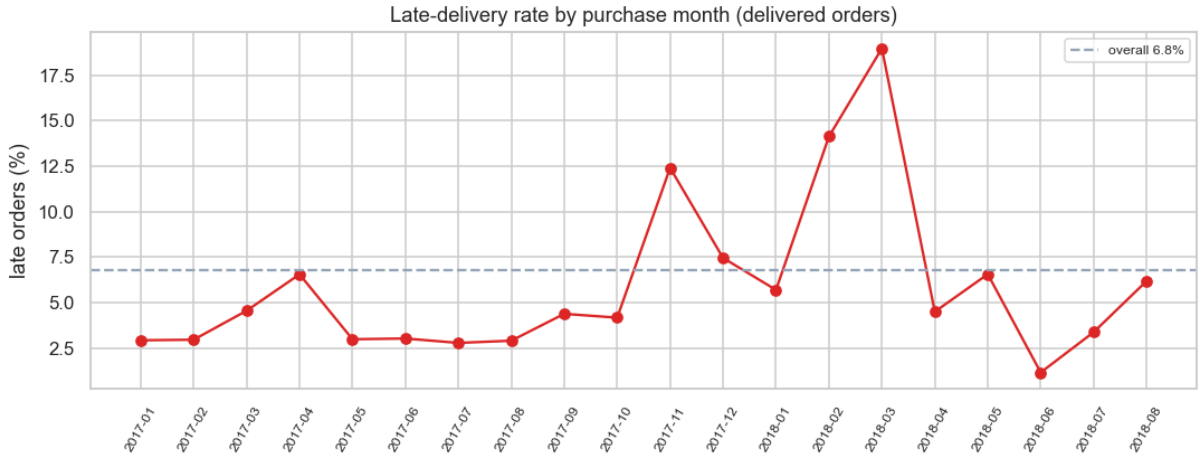
10 vs 24 days

median actual vs promised delivery time

CREW 1 INSIGHTS

Lateness depends on region and drifts over time





Worst: AL 21.5%, MA 17.5%, PI 13.9%

Best: MG 4.6%, SP 4.5%, PR 4.1%

Implication: evaluate on later months, never a random split

DATA CONTRACT

The dataset contract is the handoff

Schema: 38 columns with types, nullability, bounds and allowed values (27 states, 4 payment types, categories)

Constraints: 11 binding rules, e.g. is_late == (delivery_days > promised_days), one row per order

Target: is_late means delivered after the promised date, counted in calendar days, among delivered orders (6.8% late)

Required features: 14; forbidden post-purchase columns: 7 (no leakage)

Gate 1: the contract matches the cleaned data (21 checks)

Gate 2: every required feature exists before modeling (13 checks)

```
HANDOFF analyst->scientist
  contract_version=1.0
  contract_sha256=e881be9f2bb0
  rows=96203
  required=14 forbidden=7
```

The log line every run prints when the handoff passes; the same hash is stored in the feature spec, model metadata and model card.

CREW 2 DESIGN

Checkout-time features, time-aware evaluation

Features: 25 checkout-time features: basket, package, payment, geography (states, distance), promise and timing, category

Candidates: prior baseline, logistic regression, histogram gradient boosting (sklearn pipelines, seed 42)

Selection: time-series cross-validation on the train window, PR-AUC (imbalanced target)

Threshold: tuned on out-of-fold predictions only

Leakage guard: 7 post-purchase columns are forbidden; the features gate fails the run if one appears

split	purchase dates	orders	late rate
train	2017-01-01 to 2018-04-30	70,851	7.6%
test	2018-05-01 to 2018-07-31	19,001	3.8%
censored buffer	2018-08-01 to 2018-08-31	6,351	6.2%

August 2018 is held out as a censored buffer: the snapshot ends in mid-October 2018, so very late August orders may not have arrived yet.

RESULTS

The simpler model wins: 3x better than chance at the top

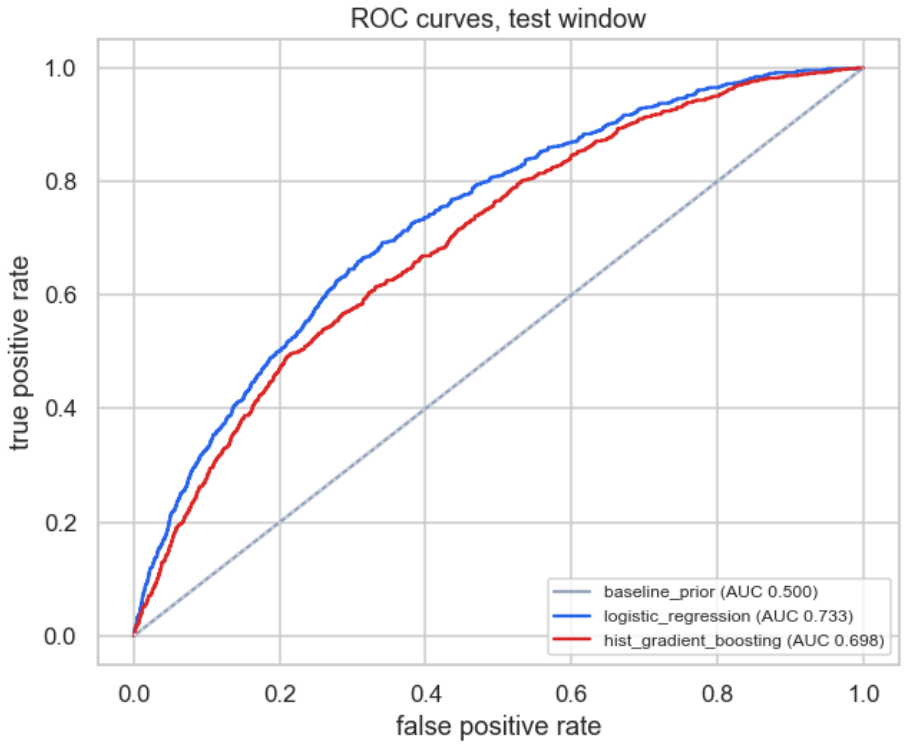
model	CV PR-AUC	test ROC-AUC	test PR-AUC	precision, top 10%
baseline prior	0.090	0.500	0.038	4.3%
hist gradient boosting	0.167	0.698	0.084	10.0%
logistic regression	**0.180**	**0.733**	**0.102**	**12.1%**

12.1%

late among the riskiest 10% (3.2x the 3.8% base rate)

0.733

test ROC-AUC (baseline 0.500)



Strongest signals: promised_days, customer_state, same_state, distance_km.

RELIABILITY

Engineered like a product: gates, failure, reproducibility

Gates: contract (21 checks), required features (13), artifacts (9); every check is logged

Graceful failure: 7 injectable faults; a failed gate writes run_status.json, exits 1 and promotes nothing

Reproducible: two fresh runs with different hash seeds give byte-identical data; the LLM run differs only in prose

Observable: pipeline.log records stages, gates, the handoff, crews, tools and every LLM call

Team workflow: one pull request per milestone, protected main, CI on every PR

```
$ olist_flow.run --no-llm --inject-fault missing_feature
FAULT INJECTED missing_feature
GATE gate_required_features required_features_present FAIL
GATE gate_required_features RESULT FAIL (11/13 checks)
RUN demo-fault FAILED at gate_required_features
run demo-fault: failed (exit 1), nothing promoted

$ olist_flow.run verify-repro
clean_data.csv        byte-identical  identical
features.csv          byte-identical  identical
metrics.json          byte-identical  identical
predictions_test.csv  byte-identical  identical
figures/*.png         same machine    11/11 identical
ALL DETERMINISTIC ARTIFACTS IDENTICAL
```

MODEL CARD

Responsible use: model card and ethics

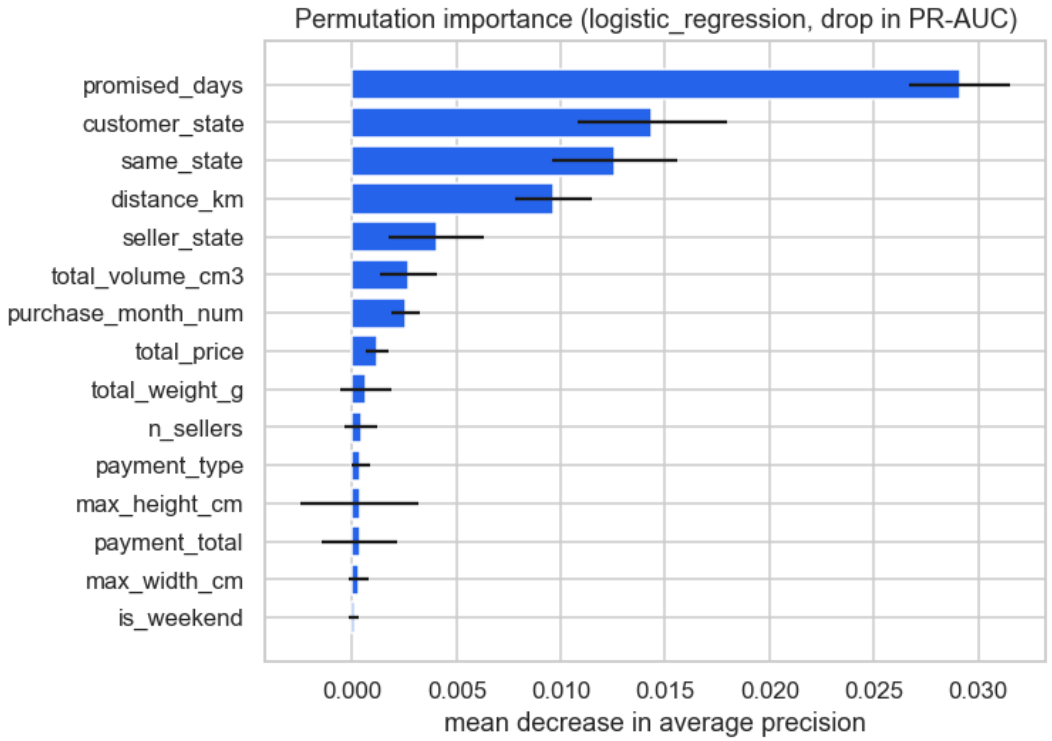
Purpose: prioritize risky orders for proactive action, never deny service or penalize customers or sellers

Limitations: delivered orders only; right-censoring; the late rate drifts, so the threshold must be re-tuned; 2017-2018 data

Regional bias: North and Northeast customers face more delays; interventions must improve their service

Privacy: pseudonymous ids; zip prefix plus timestamps are quasi-identifiers, so dashboards show aggregates only

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The score leans on the promise length and on geography, so regional fairness is part of monitoring.

DEPLOYMENT AND NEXT STEPS

From notebook to product, and what comes next

Dashboard: Streamlit app over the committed artifacts: overview, data and contract, EDA, model, predict, runs and logs

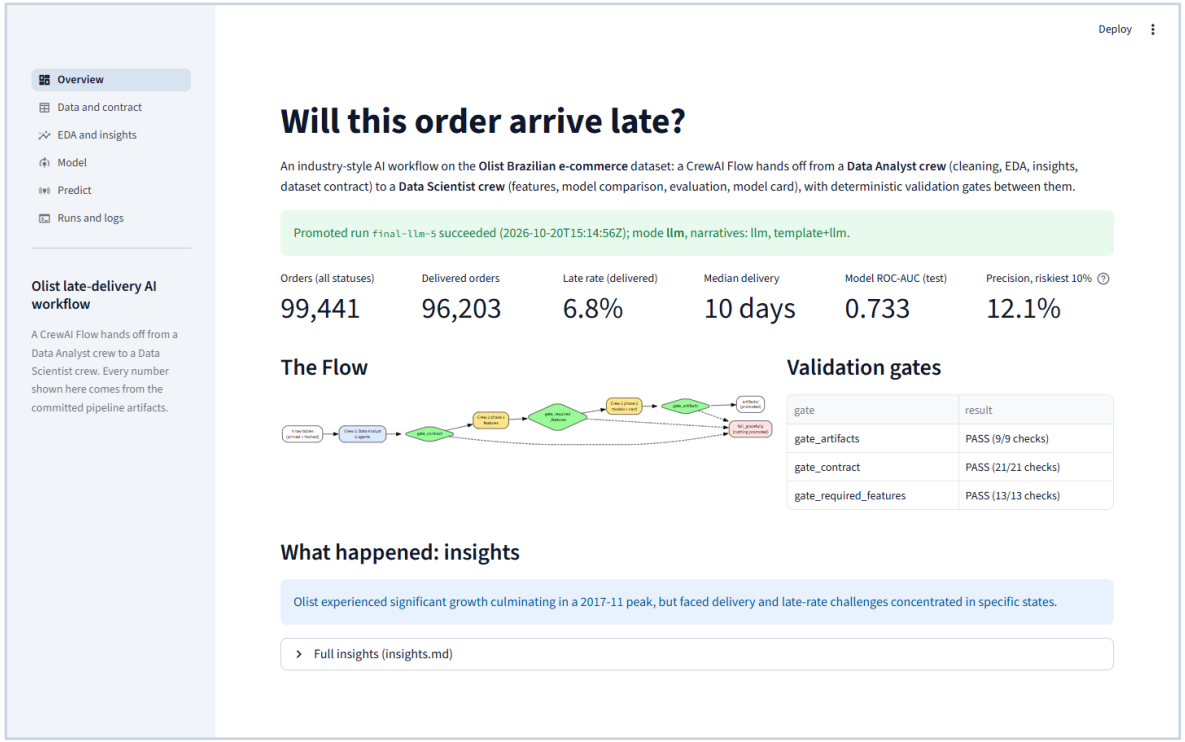
API: Flask /predict validates each order against the contract and returns the late probability and a top-10% risk flag

Run it: uv run kickoff (Gemini) or the --no-llm deterministic mode

Next: seller history features (past late rate, as of purchase time)

Next: delivery-days regression to set better promises

Next: monthly retraining with drift and calibration monitoring by state



Dashboard overview page, run final-llm-5